RAG Learning

# Theme 2.2 Macroeconomic Policy — Edexcel B

## 1. The AD/AS Model

### 1.1 Aggregate Demand (AD)

**Aggregate demand (AD):** The total planned spending on goods and services produced in an economy at a given price level, over a given time period. AD = C + I + G + (X − M).

#### Components of AD

| Component | Description | Approximate UK Share |
|---|---|---|
| **C — Consumption** | Household spending on goods and services | ~60% |
| **I — Investment** | Spending by firms on capital goods (machinery, buildings, technology) | ~17% |
| **G — Government spending** | Current and capital spending by government (excludes transfers like benefits) | ~20% |
| **(X − M) — Net exports** | Exports minus imports | Usually negative for UK (~−2%) |

#### Why the AD Curve Slopes Downward

- **Real balance (wealth) effect:** A lower price level increases the real value of people's wealth, so they spend more.

- **Interest rate effect:** A lower price level reduces demand for money, lowering interest rates, which encourages borrowing and investment.
- **Trade effect:** A lower domestic price level makes exports cheaper and imports more expensive, increasing net exports.

## Shifts of the AD Curve

Any change in C, I, G, or (X − M) that is not caused by a change in the price level will shift the AD curve. Key factors:

- **Consumer confidence and expectations** — optimism increases C.
- **Interest rates** — lower rates increase C and I (cheaper borrowing, lower returns on saving).
- **Wealth effects** — rising house/share prices make consumers feel wealthier.
- **Tax changes** — income tax cuts increase disposable income (C rises); corporation tax cuts boost I.
- **Government spending changes** — directly shifts AD via G.
- **Exchange rate** — depreciation makes exports cheaper and imports dearer, increasing (X − M).
- **Global economic conditions** — if trading partners grow, demand for UK exports rises.

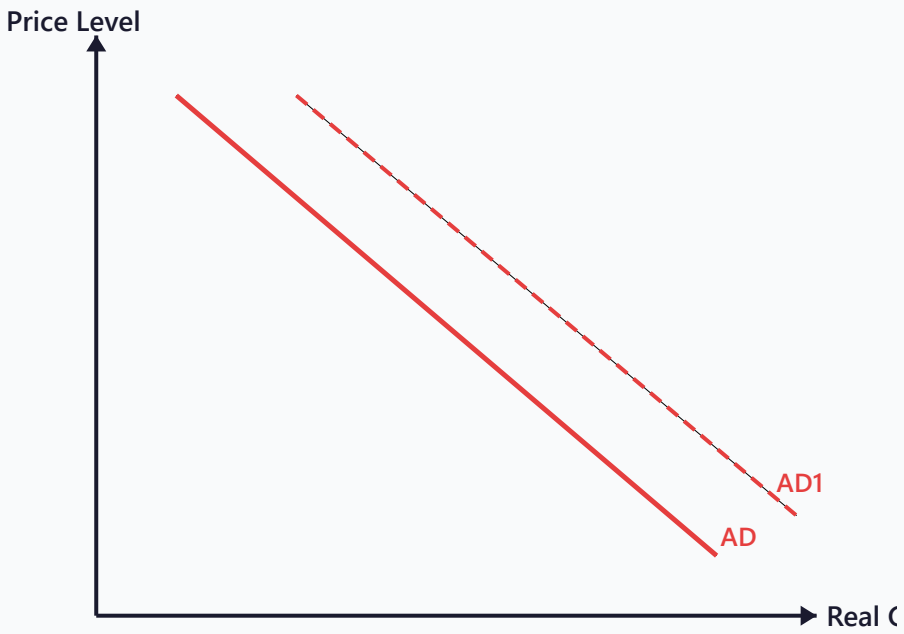


*Figure 1: An increase in aggregate demand — AD shifts to the right from AD to AD1.*

## 1.2 Aggregate Supply (AS)

**Aggregate supply (AS):** The total output of goods and services that firms in an economy are willing and able to supply at a given price level, in a given time period.

### Short-Run Aggregate Supply (SRAS)

The SRAS curve is upward sloping — as the price level rises, firms are willing to supply more because higher prices (with sticky wages in the short run) mean higher profit margins.

**Shifts of SRAS:** Changes in costs of production — wage changes, raw material prices, exchange rate changes affecting import costs, indirect taxes/subsidies, productivity changes.

### Long-Run Aggregate Supply (LRAS)

The LRAS curve is **vertical** at the full employment level of output (Yf). In the long run, output is determined by the quantity and quality of factors of production and technology — not by the price level. The economy's potential output is independent of the price level.

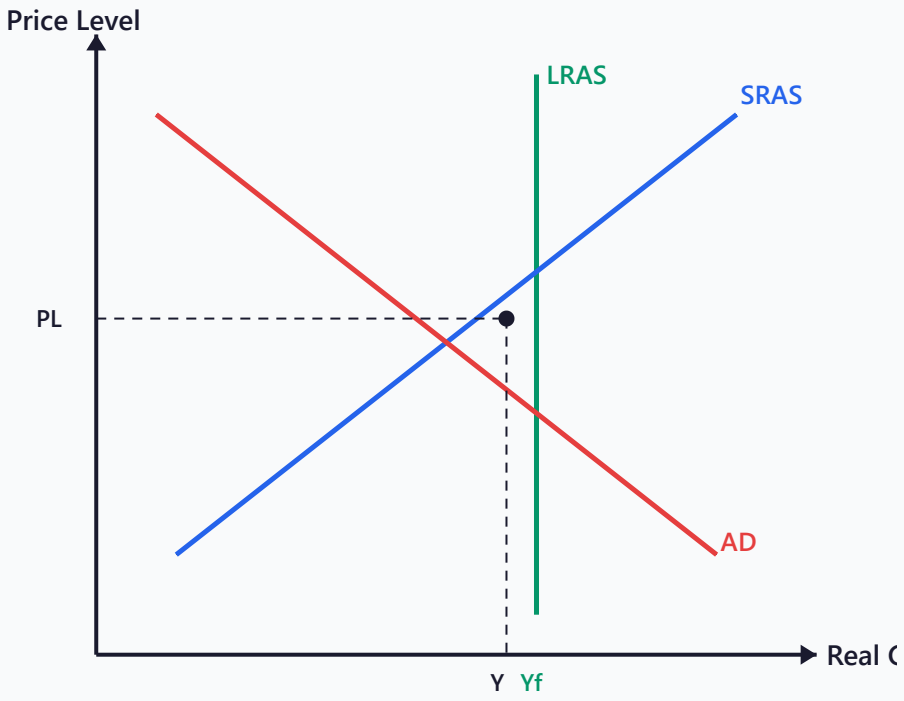

*Figure 2: The full AD/AS model with SRAS, LRAS, and AD. Macroeconomic equilibrium is where AD intersects SRAS. LRAS is vertical at full employment output Yf.*

### Shifts of LRAS

LRAS shifts rightward when the economy's productive capacity increases:

- Increase in the **quantity of labour** (immigration, population growth, higher labour force participation).
- Increase in the **quality of labour** (education, training, skills development).
- Increase in **capital stock** (investment in machinery, infrastructure, technology).
- **Technological innovation** — new methods of production.
- **Institutional improvements** — better property rights, less corruption, improved governance.
- Discovery of new **natural resources**.

## 1.3 Keynesian AS Curve

The Keynesian AS curve has three sections:

1. **Horizontal section** — lots of spare capacity; output can increase without any rise in the price level.
2. **Upward-sloping section** — as the economy approaches full capacity, bottlenecks and shortages develop, pushing up costs and prices.
3. **Vertical section** — at full employment; no further increase in real output is possible (classical range).

The Keynesian view implies that in a recession (horizontal section), demand-side policies can increase output without inflation — making fiscal expansion particularly effective.

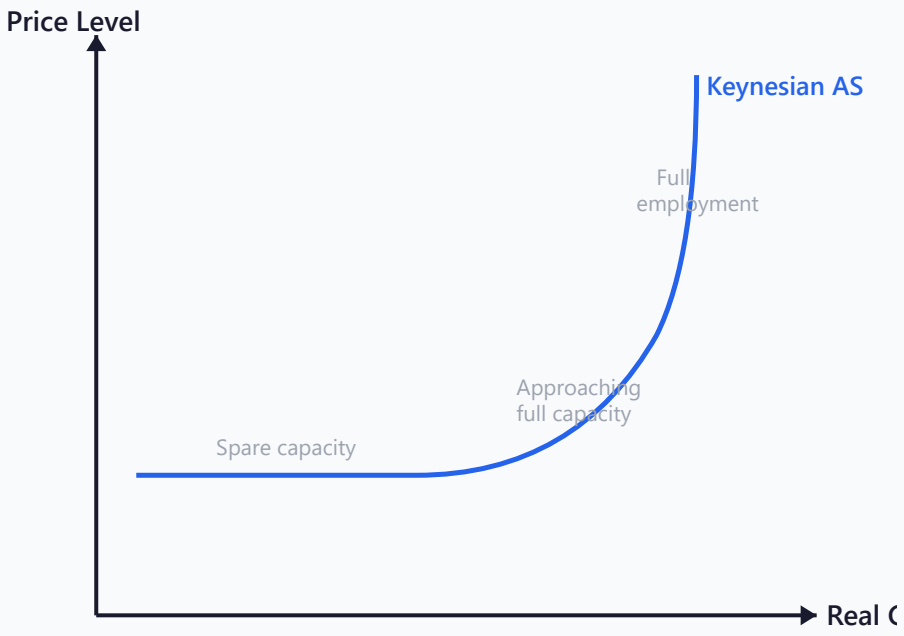


*Figure 3: The Keynesian AS curve — horizontal when there is spare capacity, becomes vertical at full employment.*

## 2. The Multiplier

**The multiplier effect:** An initial change in aggregate demand leads to a proportionally larger final change in national income. This occurs because one person's spending becomes another's income, which is then partly spent again, creating successive rounds of spending.

$$\text{Multiplier (k)} = 1 \div (1 - \text{MPC}) = 1 \div \text{MPW}$$

$$\text{MPW} = \text{MPS} + \text{MPT} + \text{MPM}$$

Where: MPC = marginal propensity to consume, MPW = marginal propensity to withdraw, MPS = marginal propensity to save, MPT = marginal propensity to tax, MPM = marginal propensity to import.

### 2.1 How the Multiplier Works

Example: The government spends an extra 1 billion on infrastructure. Construction workers receive income and spend, say, 80% of it (MPC = 0.8). Those receiving this spending also spend 80%, and so on. The multiplier = 1/(1−0.8) = 5. So the final increase in GDP is 5 billion.

## 2.2 Size of the Multiplier

The multiplier is larger when:

- MPC is high (people spend a larger share of additional income).
- MPT is low (lower tax rates mean less is withdrawn in taxes).
- MPS is low (people save less).
- MPM is low (less spending leaks to imports).

The UK multiplier is estimated to be between 1.0 and 1.5 in practice — lower than the theoretical value because of high tax rates, high import penetration, and moderate savings rates.

## 2.3 The Negative Multiplier

The multiplier works in reverse too. A fall in AD (e.g., cut in government spending) leads to a magnified fall in national income through successive rounds of reduced spending.

### Case Study: UK Austerity 2010-15

Following the 2008-09 financial crisis, the Coalition government implemented austerity — cutting public spending and raising some taxes to reduce the budget deficit. Critics (including Keynesians and the IMF initially) argued that cutting spending during a recovery would trigger a negative multiplier effect, weakening growth and possibly being self-defeating (lower growth means lower tax revenue). GDP growth was indeed sluggish from 2010-12. However, supporters argued that restoring fiscal credibility was essential to keep borrowing costs low and maintain market confidence. The debate about the size of the fiscal multiplier during this period remains one of the most contested in recent UK economic history.

### Evaluation: The Multiplier

- Assumes spare capacity exists — if the economy is at full employment, the multiplier may just lead to inflation rather than higher real output.
- Time lags — the multiplier process takes time; effects are not instantaneous.

- Crowding out — government borrowing may raise interest rates, reducing private sector investment, partially or fully offsetting the multiplier.
- Consumer and business confidence matters — if people are pessimistic, they may save extra income rather than spend it (paradox of thrift).
- Leakages to imports may be substantial in open economies like the UK.

# 3. Monetary Policy

**Monetary policy:** The use of interest rates, the money supply, and other monetary tools by the central bank (Bank of England) to influence aggregate demand, primarily to achieve the inflation target.

## 3.1 The Bank of England and the MPC

The Bank of England was granted **operational independence** in 1997. The Monetary Policy Committee (MPC) — comprising 9 members — meets 8 times per year to set the **Bank Rate** (base interest rate). The government sets the inflation target (currently **2% CPI**); the MPC decides how to achieve it.

## 3.2 How Interest Rate Changes Affect AD

The **transmission mechanism** of monetary policy:

- **Cost of borrowing:** Lower rates reduce mortgage and loan repayments, freeing up disposable income for spending. Lower rates also make business borrowing cheaper, encouraging investment.
- **Savings incentive:** Lower rates reduce the return on savings, encouraging spending over saving.
- **Asset prices:** Lower rates push up house and share prices (wealth effect), increasing consumer confidence and spending.
- **Exchange rate:** Lower rates make the UK less attractive for foreign investors, reducing capital inflows and causing depreciation. A weaker pound makes exports cheaper and imports dearer, boosting net exports.
- **Confidence:** Rate cuts signal the central bank supports growth, boosting consumer and business confidence.

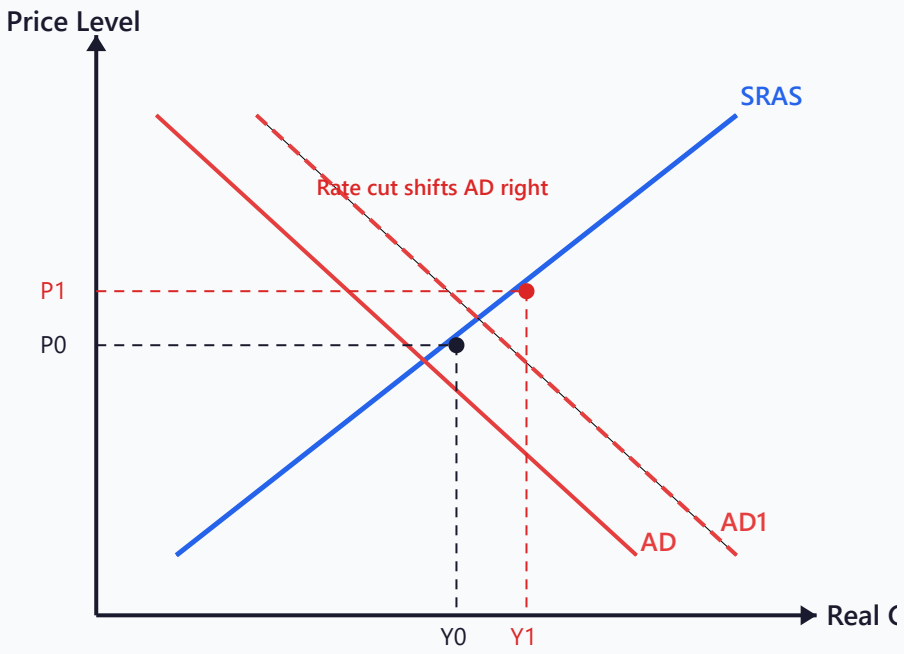


*Figure 4: Expansionary monetary policy (interest rate cut) shifts AD right, increasing real GDP and the price level.*

## 3.3 Quantitative Easing (QE)

**Quantitative easing:** An unconventional monetary policy tool used when interest rates are at or near zero (the zero lower bound). The central bank creates new electronic money to buy government bonds (gilts) and other assets from financial institutions, injecting money into the economy to lower long-term interest rates and boost lending, spending, and asset prices.

The Bank of England conducted QE from 2009 onwards, eventually purchasing 895 billion of assets. In 2022, it began **quantitative tightening (QT)** — selling bonds back to reduce the money supply and combat inflation.

**Evaluation: Monetary Policy**

- **Time lags:** Changes in interest rates take 18-24 months to have their full effect on the economy.
- **Zero lower bound:** Once rates approach zero, conventional monetary policy cannot stimulate further (requires QE).

- **Pushing on a string:** In a deep recession, cutting rates may be ineffective if banks are unwilling to lend or businesses/consumers are unwilling to borrow.
- **Blunt instrument:** Rate changes affect the whole economy — cannot target specific sectors or regions.
- **Asset price inflation:** QE may inflate house and share prices, benefiting the wealthy and worsening inequality.
- **Exchange rate effects:** Rate cuts may cause currency depreciation, which can be beneficial for exports but increases import costs and inflation.

**Case Study: Bank of England Rate Rises 2022-23**

To combat inflation that peaked at 11.1% in October 2022, the MPC raised the Bank Rate 14 consecutive times — from 0.1% in December 2021 to 5.25% by August 2023. The aggressive tightening aimed to reduce aggregate demand and bring inflation back to target. The impact was felt most through higher mortgage payments — approximately 1.4 million fixed-rate mortgage deals were due for renewal in 2023, with monthly payments increasing by hundreds of pounds. Consumer spending slowed, housing market activity fell sharply, and business investment weakened. By early 2024, inflation had fallen significantly toward the 2% target, though the economy barely avoided recession.

# 4. Fiscal Policy

**Fiscal policy:** The use of government spending and taxation to influence aggregate demand and the overall level of economic activity. Fiscal policy is set by the Chancellor of the Exchequer and announced in the Budget and Autumn Statement.

## 4.1 Types of Fiscal Policy

| Type | Action | Effect on AD | Used When |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **Expansionary** | Increase G and/or cut taxes | AD shifts right | Recession, negative output gap, high unemployment |
| **Contractionary** | Decrease G and/or raise taxes | AD shifts left | Inflationary boom, positive output gap |

## 4.2 Automatic Stabilisers

Automatic stabilisers are fiscal mechanisms that moderate economic fluctuations without deliberate government action:

- **Progressive income tax:** In a boom, rising incomes push people into higher tax bands, automatically withdrawing spending power. In a recession, falling incomes mean lower tax payments, cushioning the decline.
- **Unemployment benefits:** In a recession, spending on benefits rises automatically, maintaining demand. In a boom, benefit spending falls as employment rises.

## 4.3 The Government Budget

- **Budget deficit:** G > T — government spends more than it receives in tax revenue. The deficit must be financed by borrowing.
- **Budget surplus:** T > G — tax revenue exceeds spending. The government can repay debt.
- **National debt:** The total accumulated stock of government borrowing over time. UK national debt exceeded 100% of GDP by 2023 (over 2.5 trillion).
- **Cyclical deficit:** The part of the deficit caused by the economic cycle (lower tax receipts and higher benefit spending during recession).
- **Structural deficit:** The part of the deficit that would remain even if the economy were at full employment — reflects a fundamental mismatch between spending commitments and tax revenue.

**Evaluation: Fiscal Policy**

- **Time lags:** Fiscal policy has recognition, decision, and implementation lags — it may take months or years to design, legislate, and implement spending changes.
- **Crowding out:** Government borrowing may push up interest rates, reducing private sector investment (full crowding out would make fiscal policy ineffective). Keynesian economists argue

crowding out is minimal in a recession when there is spare capacity.

- **Political constraints:** Tax rises and spending cuts are politically unpopular — governments may favour short-term vote-winning policies over long-term economic stability.
- **Sustainability:** Large deficits lead to growing national debt and rising interest payments, which crowd out productive spending.
- **Ricardian equivalence:** Rational consumers may anticipate that government borrowing will lead to higher future taxes, so they save more now to pay for them, offsetting the stimulus (controversial theory).
- **Most effective in recession:** When there is spare capacity, crowding out is minimal and the multiplier is larger.

### Case Study: UK COVID-19 Fiscal Response

The UK government's fiscal response to COVID-19 was unprecedented. The furlough scheme (CJRS) alone cost approximately 70 billion, supporting 11.7 million jobs at its peak. Total additional fiscal spending and tax cuts related to COVID-19 exceeded 300 billion. The budget deficit reached 15% of GDP in 2020-21 — the highest since WWII. National debt surged past 100% of GDP. While this massive fiscal expansion prevented economic collapse and mass unemployment, it contributed to inflationary pressures when the economy reopened, and left the government with significantly reduced fiscal headroom for future crises.

## 5. Supply-Side Policies

**Supply-side policies:** Government policies designed to increase the productive capacity of the economy — shifting the LRAS curve to the right. They aim to improve the efficiency and flexibility of markets, increasing potential output and the long-run trend rate of growth.

### 5.1 Market-Based (Interventionist-Light) Supply-Side Policies

| Policy | How It Works | Example |
|---|---|---|

| | | |
|---|---|---|
| **Income tax cuts** | Increase incentives to work and boost labour supply | Reduction in basic rate from 20% to 19% (proposed) |
| **Corporation tax cuts** | Encourage business investment and attract FDI | UK cut corporation tax to 19% (2017), raised to 25% (2023) |
| **Deregulation** | Reduce bureaucratic burden and costs for businesses | Simplifying planning rules, reducing red tape |
| **Privatisation** | Transfer state-owned firms to private sector for greater efficiency | BT, British Gas, British Airways (1980s-90s) |
| **Trade union reform** | Reduce union power to increase labour market flexibility | Thatcher-era laws restricting strikes |
| **Welfare reform** | Reduce benefits to increase incentives to work (reduce the unemployment trap) | Universal Credit replacing multiple benefits |
| **Free trade agreements** | Remove barriers, increase competition, allow specialisation | UK-Australia FTA (2023) |

## 5.2 Interventionist Supply-Side Policies

| Policy | How It Works | Example |
|---|---|---|
| **Education and training** | Improve quality of labour (human capital), increase productivity | Apprenticeship Levy, T-Levels, university funding |
| **Infrastructure investment** | Improve transport, digital connectivity, reduce costs for firms | HS2, broadband rollout, road building |
| **R&D subsidies/tax credits** | Encourage innovation and technological progress | R&D tax credits for businesses |
| **Regional policy** | Address geographical inequality and structural unemployment | Levelling Up Fund, Enterprise Zones |

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| | | |
|---|---|---|
| **Healthcare investment** | Healthier workforce is more productive | NHS funding increases |
| **Industrial strategy** | Government actively supports key sectors | Green industrial revolution, life sciences |

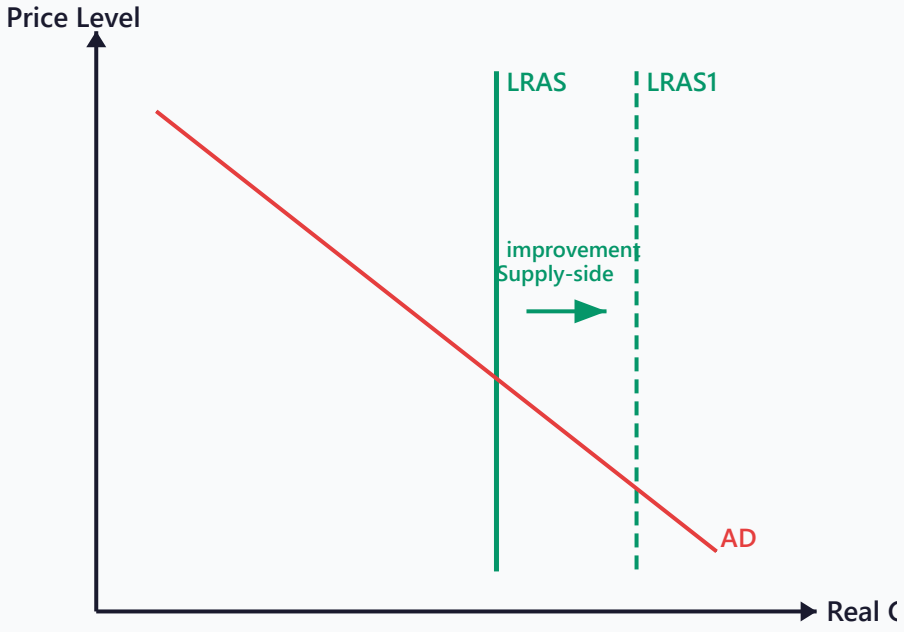


*Figure 5: Successful supply-side policies shift LRAS to the right, increasing potential output without inflationary pressure.*

**Evaluation: Supply-Side Policies**

- **Long time lags:** Education and infrastructure investment may take 5-20 years to affect productivity.
- **Costly:** Infrastructure and education require huge upfront investment — opportunity cost.
- **Uncertain outcomes:** No guarantee that spending on education will improve skills in the areas needed by employers.
- **Equity concerns:** Market-based policies (tax cuts, deregulation, welfare cuts) may increase inequality.

- **Cannot address demand-side problems:** Supply-side policies are ineffective in a recession caused by deficient demand — you need demand-side policies too.
- **Non-inflationary:** Unlike demand-side policies, successful supply-side policies increase output without increasing inflation.

## 6. Macroeconomic Policy Conflicts

The government's four main macroeconomic objectives often conflict with each other:

| Objective | Target (UK) |
|---|---|
| Economic growth | Sustained, trend rate around 2-2.5% per year |
| Low unemployment | Below the natural rate if possible |
| Low and stable inflation | 2% CPI (symmetric target) |
| Balance of payments equilibrium | Sustainable current account position |

### 6.1 Key Conflicts

- **Growth vs inflation:** Boosting AD to increase growth may cause demand-pull inflation (Phillips Curve trade-off in the short run).
- **Growth vs current account:** Faster growth increases incomes and consumer spending, much of which goes on imports — worsening the current account deficit.
- **Low unemployment vs inflation:** As unemployment falls below the NRU, labour shortages push up wages and prices.
- **Low inflation vs growth:** Contractionary policies to control inflation (higher interest rates) slow growth and may cause recession.
- **Short run vs long run:** Expansionary fiscal policy may boost growth now but create unsustainable debt that harms growth later.

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### Case Study: UK Policy Conflict 2022-23

In 2022-23, the UK faced an acute policy conflict. The Bank of England needed to raise interest rates to combat 11.1% inflation, but this risked tipping the economy into recession and increasing unemployment. Meanwhile, the government wanted to support struggling households (cost-of-living payments, energy price guarantee), which added to fiscal expansion and inflationary pressure. The September 2022 mini-budget under Liz Truss attempted large unfunded tax cuts — simultaneously expansionary fiscal policy when the Bank was tightening monetary policy. The clash spooked financial markets, causing a gilt market crisis and forcing a U-turn. This dramatically illustrated the consequences of conflicting fiscal and monetary policy.

**Exam Tip:** For any policy question, structure your answer with: (1) explain how the policy works using AD/AS diagrams, (2) analyse the short-run and long-run effects, (3) evaluate using at least 3-4 points (time lags, effectiveness depends on..., unintended consequences, conflicts with other objectives). Always draw a diagram and refer to it in your written analysis.

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